

Effects of Anthropogenic Noise Pollution on Physiological Stress Responses in the Shore Crab (*Hemigrapsus oregonensis*)

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Abstract

Anthropogenic noise pollution has become very common in coastal habitats due to activities such as shipping, construction, and shoreline recreation. Previous research has shown how noise affects fish and marine mammals but less is known about how invertebrates respond to newer recreational noise sources. We have seen that anthropogenic vibrations can cause physiological stress responses in crustaceans but the effects of newer noise sources like shoreline festivals, floating concerts, and low-frequency bass vibrations have not been investigated. It is unclear whether short-term exposure to these recreational soundscapes can cause immediate behavioral and metabolic stress responses in intertidal organisms such as the hairy shore crab, *Hemigrapsus oregonensis*. Here we show that crabs exposed to approximately 75 dB of low-frequency electronic dance music generally took longer to right themselves than control crabs, while respiration and glucose measurements showed little difference between treatments. This suggests that behavioral responses to noise could occur before there are measurable changes in respiration rate or glucose levels. Recreational noise from events such as beach parties and shoreline festivals may affect intertidal organisms but more research is needed to better understand these impacts.

Introduction

Noise pollution has become an environmental stressor as coastal communities and human infrastructure expands. These activities include pile driving, dredging, shipping noise, sonar, nearshore coastal construction, and human recreation that change the underwater sound levels (Sole et al., 2023). That change of the soundscape of the intertidal and marine environment can negatively affect marine animals that rely on sound for predator avoidance, hunting, feeding and sometimes reproduction. (Duarte et al., 2021) Although anthropogenic noise from shipping traffic and recreation boating has been widely researched with marine mammals such as orcas and whales as well as fish, not much is known about the impact of beach raves on intertidal invertebrates.

Marine invertebrates make up a large part of the nearshore ecosystems and are key contributors to predator-prey interactions and the food web. Research that was looking at the European Green Crab (*Carcinus maenas*) showed that underwater vibrations caused by anthropogenic noise pollution can cause physiological stress responses such as increased antennae beats per minute and overall activity which shows that coastal crustaceans may be more sensitive to noise than what has been previously thought. (Aimon et al., 2021) The hairy shore crab, *Hemigrapsus oregonensis*, is a Pacific Ocean intertidal species that lives in shallow coastal habitats where human activity is common and are a useful species for studying how noise pollution can affect marine life. Previous work that looked at shipping noise tolerance in *H. oregonensis* showed that this species may be able to tolerate some forms of anthropogenic sound exposure (Birch et al., 2025). Different

noise sources do vary in intensity, duration, and frequency from boat noise to coastal construction such as seawalls which can make it difficult to predict how intertidal organisms will respond to newer recreational disturbances like shoreline festivals or floating concerts.

Those new events such as floating concerts, shoreline festivals, and recreational beach raves continue to increase in popularity and generate intense sounds and vibrations that travel through shallow coastal environments. Looking at respiration rate, righting time, and glucose levels in *Hemigrapsus oregonensis* could help us understand how recreational noise pollution influences physiological stress responses in intertidal crabs and how increasing human activity may affect intertidal ecosystems.

Methods

Study Organism and Experimental Design

Twelve hairy shore crabs (*Hemigrapsus oregonensis*) were used to examine whether acute anthropogenic noise exposure induced physiological stress responses. Crabs were randomly assigned (sex and size) into either a Noise treatment (experimental, n=6) group or a Quiet treatment (control, n=6) group. Experimental crabs were exposed to approximately 75 dB of music using a speaker placed next to the experimental tank for two hours once per week. Control crabs were maintained under identical conditions without EDM exposure. Noise exposure consisted of low-frequency electronic dance music (EDM) and techno music intended to simulate recreational noise sources such as shoreline festivals, floating concerts, and beach events. Glucose assay was performed during week three after two hours of exposure via hemolymph extraction to account for possible fatal hemolymph extractions. Two tanks with lids

were used to maintain treatment conditions. Hairy shore crabs were individually identified using superglued tags. Mortality and molting events were recorded when observed. Two crabs died during molting between Weeks 1 and 2 and were excluded from subsequent analyses.



Figure 1. Tank setup during 0:00-1:00 hours. The experimental crabs were exposed to music throughout the duration of the provided time. The quiet crabs were placed in an isolated room.

Respiration Assay

Respiration response was measured using a resazurin assay for metabolic activity. Individual crabs were placed into separate cups containing 8 mL of resazurin working solution after 1:00.

At the beginning of the assay ($T_0 = 0:00$), 200 μL aliquots were collected from each chamber and transferred into a 96-well plate. The crabs for the treatment group were exposed to EDM music for 1:00 hr in their respective tanks while the control group remained in a quiet room. Following 1:00 hr, we extracted both the treatment and the control crabs and placed them in a 6-well plate while still being exposed to music for the experimental group. At 1:30, a 200 μL aliquot ($T_1 = 1:30$) was collected from each crab well and transferred to a 96-well plate for each crab.

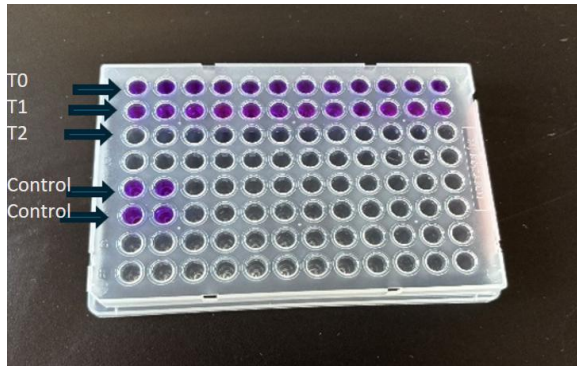


Figure 2. 96-well plate after aliquots extracted at time 1:30. The wells at E1 and E2 were Quiet control and F1 and F2 were Noise control aliquots.

At 2:00, a final aliquot (T2=2:00) was collected and transferred into the 96-well plate for all crabs. The experimental group was exposed to music for the full two hours. Fluorescence measurements were quantified using a spectrophotometer plate reader.



Figure 3. Experimental crab setup during 1:00-2:00 hours. The experimental crabs were exposed to music throughout the duration of the provided time. Aliquots taken at 1:30 and 2:00.

Righting Time

Righting time was measured as a behavioral indicator of stress response. Individual crabs were placed upside down on their carapace and the time required to return to an upright position was recorded. Longer righting times were interpreted as a potential indicator of altered physiological or behavioral performance following noise exposure. One crab in the Control group molted in the 6-well plate and therefore had no carapace for righting time after

the resazurin assay during week 2 and was excluded from righting time measurements.

Glucose Assay

Hemolymph glucose concentration was measured using a glucose colorimetric assay kit. Hemolymph samples were collected from individual crabs ($n = 10$) using a 100 μL insulin syringe and placed in storage tubes. Sample size was reduced to 10 crabs due to two molting mortalities in between Week 1 and Week 2. Absorbance values were compared between treatment groups after that.

Statistical Analysis

Respiration response, glucose absorbance, and righting time were compared between Noise and Quiet treatment groups. Outliers were documented during analysis and excluded from figure visualization when necessary to improve interpretation of overall treatment trends. Data was collected in Google Sheets and then analyzed in R and Microsoft Excel.

Results

Respiration assays (Fig. 4 & Fig. 5) did not show strong differences between control and treatment crabs despite the visual trends between groups. With p -values greater than 0.05 from $T0 = 0.91$, $T1 = 0.57$ and $T2 = 0.54$, there was no statistically significant difference between the experimental and control groups. When averaging all three weeks together, control (quiet) crabs showed higher respiratory responses than treatment (noise) crabs. Oxygen demand also increased over time in both groups and appeared non-linear throughout the assay. Both treatment and control crabs showed larger increases in respiration during the second 30-minute period (1:30-2:00) compared to the first 30 minutes (1:00-1:30).

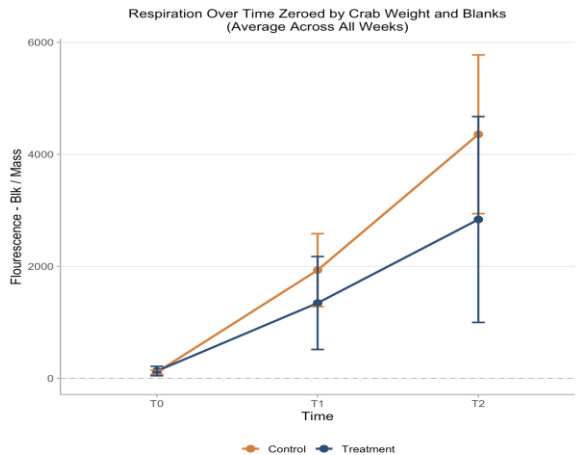


Figure 4. Fluorescence measurements from the resazurin respiration assay across all three weeks for the Quiet (control) and Noise (experimental) treatment groups at T0 (0:00), T1 (1:30 hr), and T2 (2 hr).



Figure 5. Weekly fluorescence measurements from the resazurin respiration assay for Quiet (control) and Noise (experimental) treatment groups at T0 (0:00), T1 (1:30 hr), and T2 (2 hr).

Righting time (fig. 6) measurements showed differences between treatment groups. Crabs exposed to noise generally took longer to return to an upright position following inversion compared to crabs in the Quiet (control) group. The average righting time for the Quiet (control) group was 0.995 seconds, while the average righting time for the Noise (experimental) group excluding extreme values was 1.193 seconds, representing an increase of 19.9% (0.198 seconds) relative to the control (quiet) crabs. Even though noise-exposed crabs had longer righting times on average, the differences between groups were still relatively small.

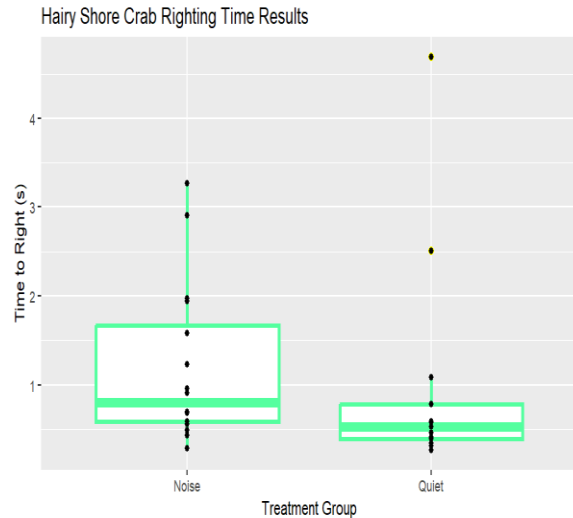


Figure 6. Crab righting time after 2:00 of either noise exposure or quiet. Righting time measurements for Quiet (control) and Noise (experimental) crabs across all three weeks ($n = 31$). Noise-exposed crabs generally showed longer righting times than control crabs.

Two extreme observations were identified within the Noise treatment group (46.81, 56.24 seconds). These values were excluded from figure visualization because they compressed the remaining data and reduced the plot visibility but were kept for calculation purposes wherever appropriate. When these extreme observations were included, the average righting time for noise-exposed crabs increased to 6.785 seconds. We originally expected the noise treatment to increase stress responses across respiration, glucose concentration, and righting time. Righting time did show differences between treatment groups, but respiration response and glucose absorbance measurements were both negligible.

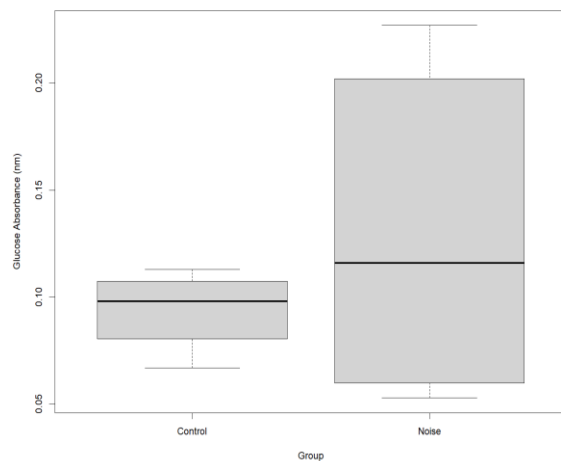


Figure 7. Hemolymph glucose absorbance between the Quiet (control) and Noise (experimental) groups during Week 3. Even though the averages are similar, the noise group shows a much higher variation.

Discussion

One of the more surprising findings was that control crabs visually showed greater respiration response than treatment crabs despite exposure to approximately 75 dB of anthropogenic sound. We expected noise exposure to increase energetic demand and respiration due to stress response or altered activity and more behavior. However, we observed no clear differences despite the visual trends seen. One explanation may be that *H. oregonensis* is able to tolerate some forms of short-term noise exposure. Birch et al. (2025) found that *H. oregonensis* showed little to no response to the shipping noise which could help explain why strong metabolic differences were not observed in this study. Another possibility is that two hours of noise exposure was not enough time to produce measurable physiological changes in respiration. Future research could focus on longer exposure to noise over days to weeks to see any chronic responses to noise pollution.

The strongest response that we observed came from righting time measurements. Crabs exposed to EDM music generally took longer to

return to an upright position after inversion than the control crabs. The differences between groups remained relatively small overall but the trend was consistent. The increased righting times may reflect short-term behavioral disruption or disorientation after noise exposure from the low frequency bass and vibration of EDM music.

Glucose absorbance measurements also produced weaker responses than what was originally predicted. Noise exposed crabs showed greater variation in absorbance values compared to control crabs, but there was no clear pattern. One possible explanation is individual variation between crabs. Stress responses likely differed among individual crabs depending on the molt stage, size, sex and tolerance to disturbance. Glucose response may present differently than behavioral measurements and may not have been fully captured under their short term exposure conditions.

There were also limitations to this experiment. Sample size remained relatively small ($n = 12$), and crab mortality reduced sample sizes during later weeks ($n = 10$). The control (quiet) crabs also consumed the crabs that molted and died while the experimental group did not have any molting mortalities that were left in the tank. The dead crabs also affected the water quality, which may have also affected the behavior and stress response of the control crabs for week two. Molting events also complicated measurements for righting time observations. One control crab molted during the resazurin assay and could not be included for righting time due to lacking a hardened carapace. Another limitation or constraint was that the control crabs were still experiencing some noise in a separate room as it was by a door and hallway, but not to the extent that the experimental group was experiencing both sound and vibration. Future work could increase their sample size, increase

or decrease exposure duration and investigate chronic noise exposure.

Although the physiological responses measured in this study were inconsistent, noise exposure appeared to affect behavioral responses more than metabolic responses in *H. oregonensis*. Crabs exposed to low-frequency EDM and techno music generally showed longer righting times than control crabs even though respiration and glucose measurements showed little overall difference between treatment groups. The longer righting times suggest that behavioral responses to noise may appear before larger physiological responses become obvious. Recreational noise produced by events such as raves and beach parties could affect intertidal organisms but more research is needed to understand these issues.

References

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